Solution Requirements

Date	1 July 2026
Team ID	SWTID-2026-4523
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

● Functional Requirements (FR) define specific behaviours, functions, or features of the system (What the system should do).

● Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should be).

Step 1: Functional Requirements (FR)

S.NO	Requirement Category	Requirement Description	Priority
1	Authentication	Users (bank staff/admin) must securely log in using a username and password before accessing the system	High
2	Authorization Levels	Admin can manage users, datasets, and models. Staff can only enter applicant details and view prediction results.	High
3	External Interfaces	The system accepts applicant data through a web interface and uses the trained Machine Learning model for prediction.	Medium
4	Transaction Processing	The system processes applicant details, performs preprocessing, applies the ML model, and predicts whether the credit card application is Approved or Denied	High
5	Reporting	Generate prediction results with applicant details and allow viewing of previous prediction records.	Medium
6	Business Rules	Credit approval is based on the trained ML model using applicant information such as income, age, employment status, family status, and credit history.	High
7	Compliance to Laws or Regulation	User information must be kept confidential and handled according to applicable data privacy standards.	High

8	Other	The system should validate user inputs and display meaningful error messages for invalid or missing data.	Medium

Step 2: Non-Functional Requirements (NFR)

S.NO	NFR Category	Requirement Description	Target Metric/Acceptance Criteria
1	Performance & Speed	Prediction results should be generated quickly after user submission.	Response time less than 2 seconds
2	Scalability	The system should support increasing numbers of users and prediction requests.	Supports 1000+ prediction requests per day
3	Security & Data Privacy	Applicant data must be securely stored and transmitted with proper authentication.	Password-protected login and encrypted data storage
4	Reliability & Availability	The application should remain available with minimal downtime.	99% uptime
5	Usability & Accessibility	The interface should be simple, user-friendly, and easy to understand.	New users should complete a prediction without training
6	Other (Maintainability)	The system should allow easy model updates and future enhancements without major code changes.	Modular code with easy maintenance